

Statistics

May 5, 2026

Kexing Ying

1 Regression function

For square loss function $L(Y, f(X)) = (Y - f(X))^2$, the expected prediction error is

$$\text{EPE}(f) = \mathbb{E}[(Y - f(X))^2].$$

It is easy to see that the regression function $f(x) = \mathbb{E}[Y \mid X = x]$ minimizes the expected prediction error. Indeed, as all $\sigma(X)$ -measurable random variables are of the form $f(X)$, it is equivalent to minimize $\|Y - Z\|_2$ over all $\sigma(X)$ -measurable random variables Z . This is of course achieved by the orthogonal projection of Y onto the closed subspace of $\sigma(X)$ -measurable functions which is exactly $\mathbb{E}[Y \mid X]$.

The nearest neighbour method directly tries to estimate this regression function. In particular, intuitively, given iid $(X_i, Y_i)_{i=1}^n$ and x , we hope to approximate

$$\mathbb{E}[Y \mid X = x] \approx \frac{1}{k} \sum_{i=1}^n \mathbf{1}_{\{\|X_i - x\| \leq \|X_{(k)} - x\|\}} Y_i.$$

Namely, we take the sample average of the Y_i values corresponding to the k nearest neighbours of x . By taking a variance-bias decomposition, if we take k, n in such a way that $k \rightarrow \infty$ and $\frac{k}{n} \rightarrow 0$, then under some mild assumptions on the distribution of (X, Y) , we have the approximation converges in the limit.

The nearest neighbour method (and local methods in general) is problematic in high dimensions. For fixed x , if we want our estimate to depend on the data within a neighbourhood of x which captures a fixed proportion of the data, then the radius of this neighbourhood has to grow exponentially in the dimension (so rather contradicts the intuition of locality). This is called *the curse of dimensionality*.

A natural extension of the nearest neighbour method is, instead of taking the average of the nearest neighbours, we can take a weighted average of all points weighted by their distance to x . We call the weight function K the kernel function and a simple estimate (Nadaraya–Watson) is

$$f(x) = \frac{1}{Z} \sum_{i=1}^n K(x, X_i) Y_i$$

with the normalisation $Z = \sum_{i=1}^n K(x, X_i)$.

2 Linear regression

Let us consider the OLS estimator from a probabilistic perspective. Suppose $\mathcal{H}, \mathcal{K}$ are two Hilbert spaces and we have random variables $X \in \mathcal{H}$ and $Y \in \mathcal{K}$. The fundamental assumption one makes with linear regression is that the regression function is linear, i.e. there exists a linear operator $\beta : \mathcal{H} \rightarrow \mathcal{K}$ such that $\mathbb{E}[Y \mid X] = \beta X$ (we assume w.l.o.g. that both X, Y are centred).

Now, recalling that the covariance operators C_X and C_{XY} are defined by

$$C_X : \mathcal{H} \rightarrow \mathcal{H}, \quad C_X f = \mathbb{E}[\langle f, X \rangle_{\mathcal{H}} X]$$

and

$$C_{XY} : \mathcal{H} \rightarrow \mathcal{K}, \quad C_{XY} f = \mathbb{E}[\langle f, X \rangle_{\mathcal{H}} Y].$$

Thus, we compute

$$\langle C_{XY} h, k \rangle = \mathbb{E}[\langle X, h \rangle \langle Y, k \rangle] = \mathbb{E}[\langle X, h \rangle \langle \mathbb{E}[Y | X], k \rangle] = \mathbb{E}[\langle X, h \rangle \langle \beta X, k \rangle] = \langle \beta C_X h, k \rangle$$

for all $h \in \mathcal{H}$ and $k \in \mathcal{K}$. Hence, we have $C_{XY} = \beta C_X$. Thus, in the case where C_X is invertible, we have $\beta = C_{XY} C_X^{-1}$.

In the finite-dimensional case, the OLS estimator is exactly the empirical version of this formula. In particular, suppose now $\mathcal{H} = \mathbb{R}^p$ and $\mathcal{K} = \mathbb{R}$. Then, we have the covariance matrices $C_X = \mathbb{E}[X X^\top]$ and $C_{XY} = \mathbb{E}[X Y^\top]$. Now, given iid data $(\mathbf{X}, \mathbf{Y}) = (X_i, Y_i)_{i=1}^n \in \mathbb{R}^{n \times p} \times \mathbb{R}^n$, we can estimate C_X and C_{XY} by their empirical versions

$$\hat{C}_X = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top = \frac{1}{n} \mathbf{X}^\top \mathbf{X}, \quad \hat{C}_{XY} = \frac{1}{n} \sum_{i=1}^n X_i Y_i^\top = \frac{1}{n} \mathbf{X}^\top \mathbf{Y}.$$

Thus, the OLS estimator is exactly $\hat{\beta} = \hat{C}_{XY} \hat{C}_X^{-1}$. Note that the OLS estimator one typically writes down is the transpose of what we have here as one typically instead assumes $Y = X\beta + \varepsilon$.

3 Ridge regression and kernels

The OLS estimator is the unbiased estimator minimising MSE. However, it is not the estimator minimising MSE among all estimators. In particular, via the bias-variance decomposition, one might hope to reduce the MSE by introducing some bias. The ridge regression estimator achieves this by penalising large values of β . This is intuitive as β is large when C_X is close to singular and thus the OLS estimator is unstable. The ridge regression estimator instead optimises

$$\hat{\beta} = \arg \min_{\beta} \left(\frac{1}{n} \sum_{i=1}^n (Y_i - X_i^\top \beta)^2 + \lambda \|\beta\|^2 \right)$$

for some regularisation parameter $\lambda > 0$.

3.1 The Cameron–Martin space and RKHS

Suppose now X is a Gaussian random variable in a Hilbert space $\mathcal{B}$ with law μ . We recall that the Cameron–Martin space $\mathcal{H}_\mu$ of μ is defined as the closure of the pre-Cameron–Martin space

$$\mathcal{H}_\mu^\circ = \{x \in \mathcal{B} : \exists x^* \in \mathcal{B}^*, l(x) = C_\mu(l, x^*) \forall l \in \mathcal{B}^*\} = C_\mu(\mathcal{B}^*)$$

where in the latter equality we identify $C_\mu : \mathcal{B}^* \rightarrow \mathcal{B}$ by $C_\mu l = \int l(x) x \mu(dx)$ with the integral understood in the Bochner sense.

We show that the pre-Cameron–Martin space is exactly the RKHS (in the statistician’s sense) associated to the kernel $k(h, k) = C_\mu(h, k) = \int h(x) k(x) \mu(dx)$ (to be more precise, the RKHS of k is the canonical identification of $\mathcal{H}_\mu^\circ$ in $\mathcal{B}^{**}$).

Recall that the RKHS $\mathcal{H}$ associated to a kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a Hilbert space of functions $f : \mathcal{X} \rightarrow \mathbb{R}$ such that $k(x, \cdot) =: k_x \in \mathcal{H}$ for all x and

$$\langle f, k(x, \cdot) \rangle_{\mathcal{H}} = f(x).$$

In this case, $\mathcal{X} = \mathcal{B}^*$ and $\mathcal{H} = \mathcal{H}_\mu$ with the canonical identification $x \in \mathcal{H}_\mu \subseteq \mathcal{B}^{**}, x : l \mapsto l(x)$. Indeed, in this case, $k(l, \cdot) = C_\mu l$ and for any $x \in \mathcal{H}_\mu$, we have that

$$\langle x, k(l, \cdot) \rangle_\mu = \langle x, C_\mu l \rangle_\mu = C_\mu(x^*, l) = l(x) = x(l)$$

where we used the fact that $(C_\mu l)^* = l$.